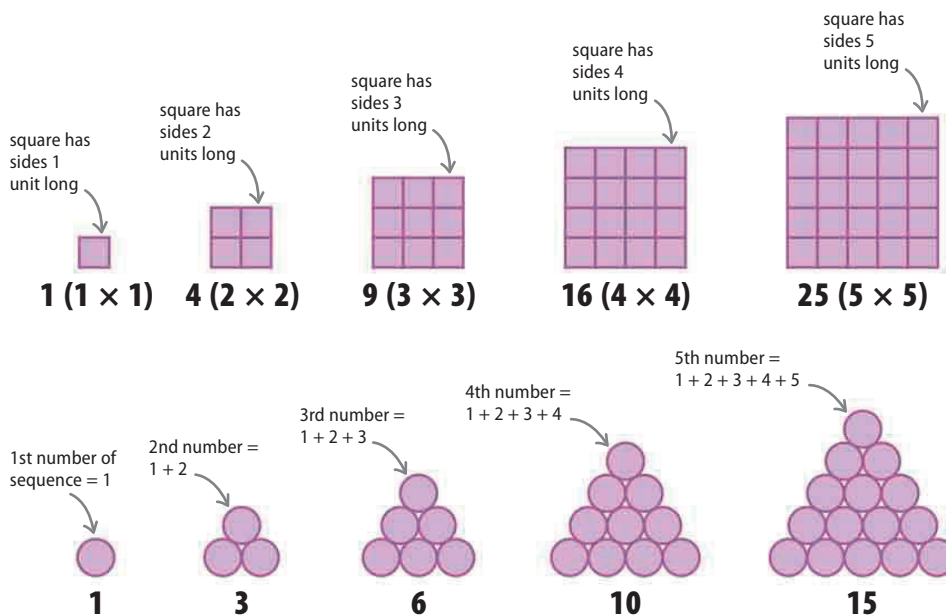


Sequences

A sequence is a series of numbers written as an ordered list where there is a particular pattern or “rule” that relates each number in the list to the numbers before and after it. Examples of important mathematical sequences are shown below.



◁ Square numbers

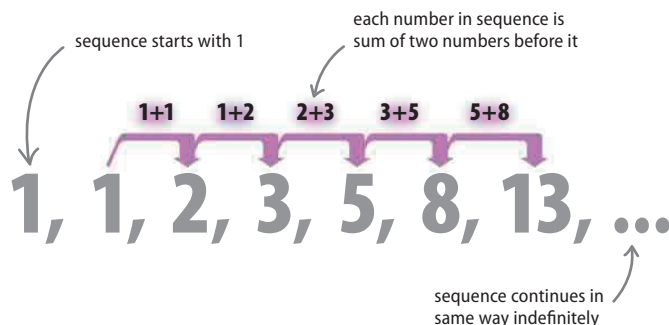
In a sequence of square numbers, each number is made by squaring its position in the sequence, for example the third number is 3^2 ($3 \times 3 = 9$) and the fourth number is 4^2 ($4 \times 4 = 16$).

◁ Triangular numbers

In this sequence, each number is made by adding another row of dots to the triangular pattern. The numbers are also related mathematically, for example, the fifth number in the sequence is the sum of all numbers up to 5 ($1 + 2 + 3 + 4 + 5$).

Fibonacci sequence

Named after the Italian mathematician Leonardo Fibonacci (c.1175–c.1250), the Fibonacci sequence starts with 1. The second number is also 1. After that, each number in the sequence is the sum of the two numbers before it, for example, the sixth number, 8, is the sum of the fourth and fifth numbers, 3 and 5 ($3 + 5 = 8$).



Pascal's Triangle

Pascal's triangle is a triangular arrangement of numbers. The number at the top of the triangle is 1, and every number down each side is also 1. Each of the other numbers is the sum of the two numbers diagonally above it; for example, in the third row, the 2 is made by adding the two 1s in the row above.

